Stores an exception in the asynchronous result associated with `*this`.

Declaration

```
void set_exception(std::exception_ptr e);
```

Preconditions

`*this` has an associated asynchronous result. `(bool)e` is `true`.

Effects

Stores `e` in the asynchronous result associated with `*this`.

Postconditions

The asynchronous result associated with `*this` is *ready* with a stored exception. Any threads blocked waiting for the asynchronous result are unblocked.

Throws

An exception of type `std::future_error` with an error code of `std::future_errc::promise_already_satisfied` if the asynchronous result already has a stored value or exception.

Synchronization

Multiple concurrent calls to `set_value()` and `set_exception()` are serialized. A successful call to `set_exception()` happens-before a call to `std::future<Result-Type>::get()` or `std::shared_future<ResultType>::get()`, which retrieves the exception stored.

Std::Promise::Set_Exception_At_Thread_Exit Member Function